AI & Cybersecurity – Day 2

INTRODUCTION TO GEN - AI

- Aakash K

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What is the first word that comes to your mind when we say *Gen-AI* ?

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Event code
XAXUUK

https://app.wooclap.com/events/XAXUUK/live-session

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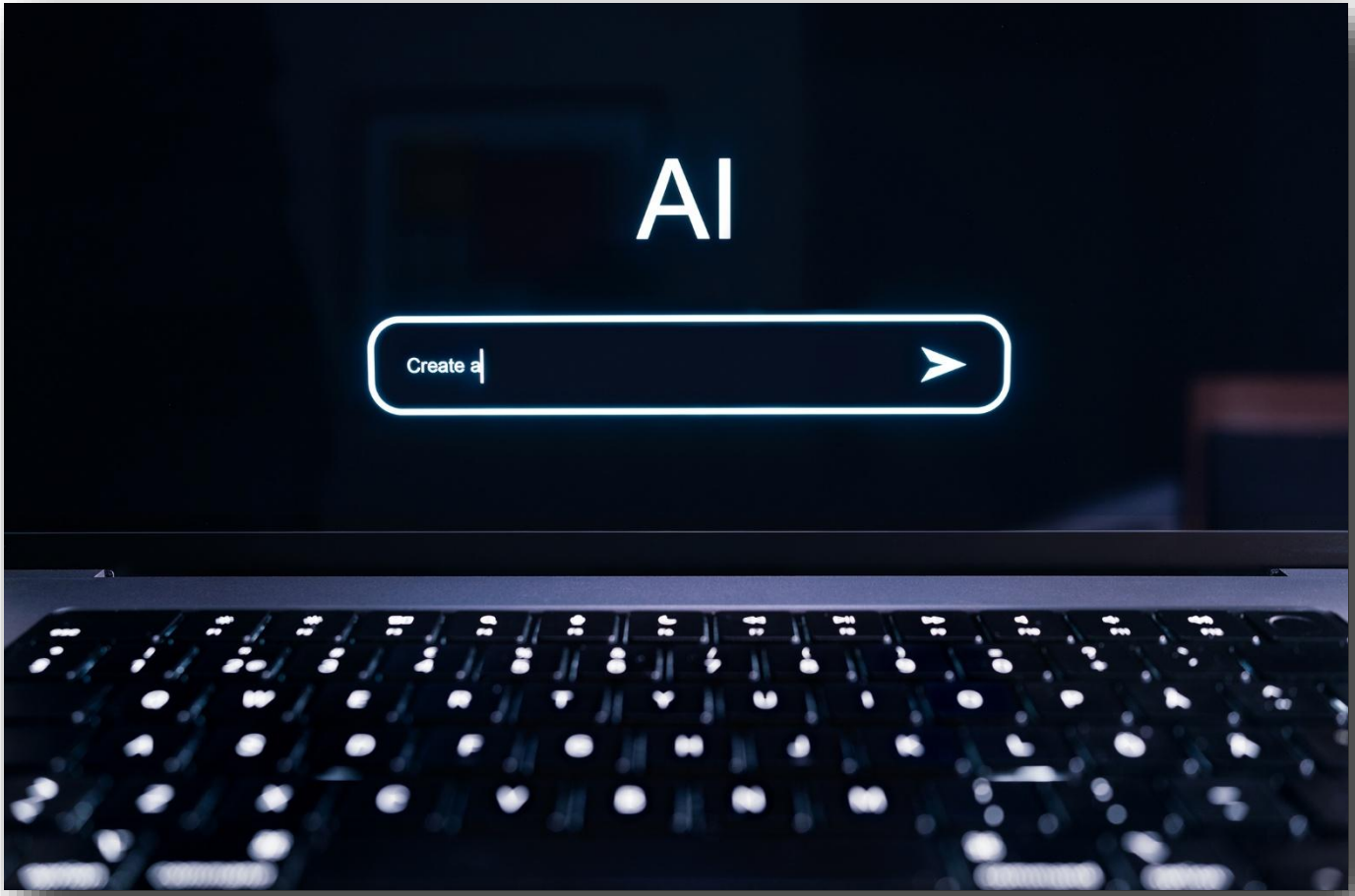
What is GenAI?



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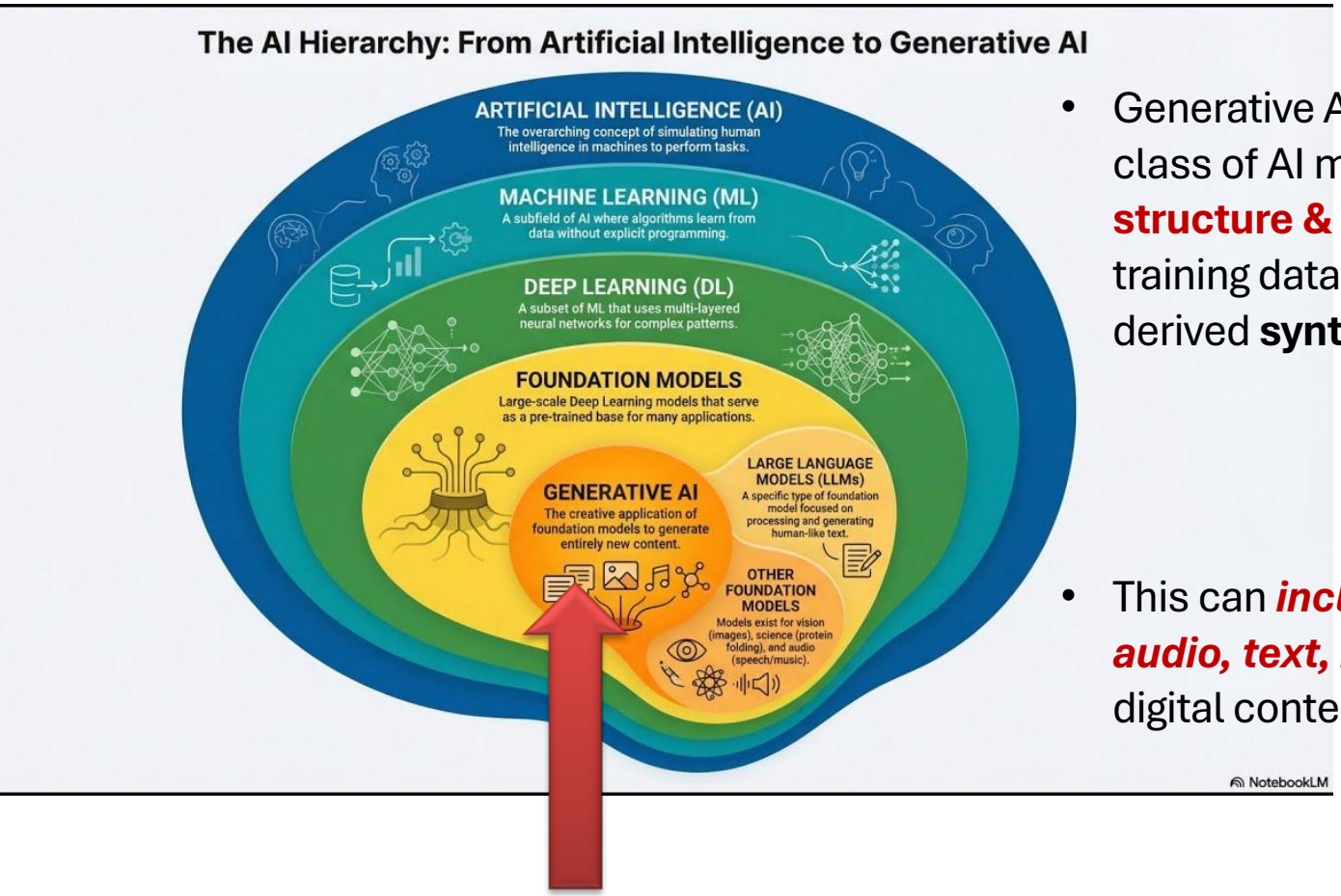
What is GenAI ?



- Generative AI (GenAI) refers to "the class of AI models that emulate the **structure & characteristics** of training data in order to **generate** derived **synthetic** content.
- This can ***include images, videos, audio, text, images, code*** and other digital content."

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GenAI Landscape

Conversational AI & Associated Large Language Models (LLMs)

- ChatGPT: o3, GPT 4.5, GPT 4.1
- Gemini: Gemini 2.5 Pro, Gemini 2.5 Flash, Gemini 2.0 Flash
- Claude: 3.7 Sonnet, 3.5 Sonnet, 3 Opus
- Grok: Grok 3, Grok 3 Mini, Grok 2
- Meta AI: LLaMA 4 Maverick, LLaMA 3.3 49B, LLaMA 3.1 70B
- deepseek: DeepSeek V3, DeepSeek R1, DeepSeek V2.5

AI Presentation & Design

Gamma, Adobe Firefly, Canva, beautiful.ai

AI Avatars

tavus, HeyGen, synthesia

Text-to-Speech

Eleven Labs, Google Cloud Text to Speech, MURF.AI, Text-to-Speech

AI Image Generation

GPT 4o IMAGE GENERATION, Gemini Imagen 3, Midjourney, stability.ai

AI Video Creation

Google Veo 2, Sora, runway, LUMA AI

Meeting Transcription

Otter.ai, Fireflies, FATHOM, zoom AI Companion

AI Collaboration, Productivity, & Enterprise Search

Notion, glean, coveo, Amazon Kendra, ATLASSIAN Intelligence, slack AI, grammarly

AI Search

Google AI Mode, ChatGPT Search

AI Research Summarization

Gemini Deep Research, Deep research, NotebookLM, Elicit

Vector Databases & Optimization

Pinecone, redis, [vectorize], mongoDB. Atlas, Amazon OpenSearch Service, pgvector

AI Agent Development

LangChain, LlamaIndex, crewai, AutoGen

Agentic Automation & Orchestration

n8n, appian, UiPath, zapier, workato

AI Platforms

Vertex AI, Azure AI Foundry, Amazon Bedrock, Hugging Face, NVIDIA AI Enterprise, Amazon SageMaker

Enterprise Platforms

Agentforce, servicenow, Microsoft Copilot Studio, SAP AI Core

AI Coding & Development

CURSOR, bolt.new, replit, GitHub Copilot, Windsurf, manus

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As of May 2025

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Topic - Generative AI

- Concepts & Terminologies
- Risks & Mitigations
- Hands on Workshop

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Learning Outcome

- At the end of this module, learners should be able to ***understand*** typical ***GenAI terminology*** and ***concepts***.

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What is a Foundation Model ?

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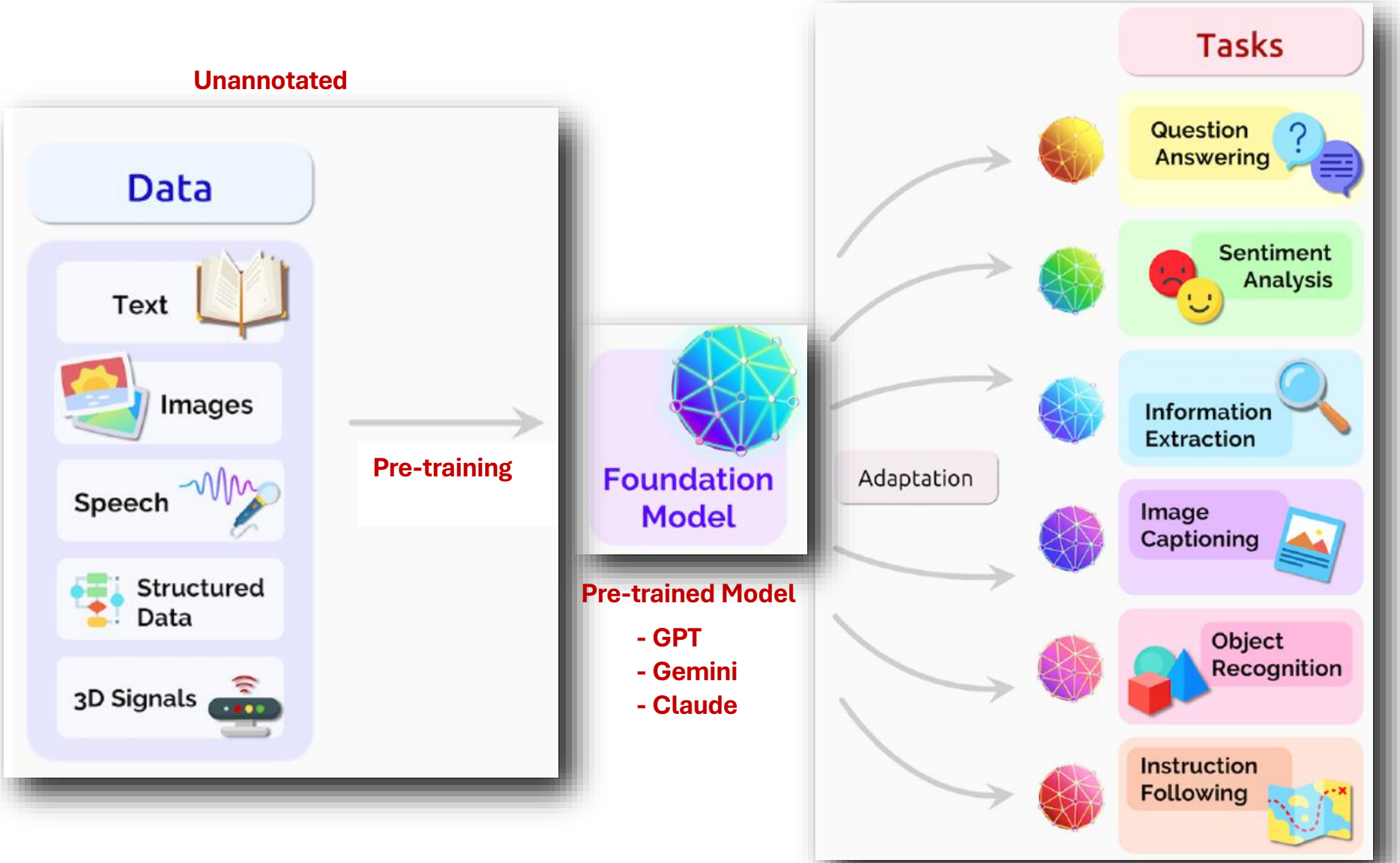
Foundation Model



Ref: Video at https://hai.stanford.edu/news/what-foundation-model-explainer-non-experts

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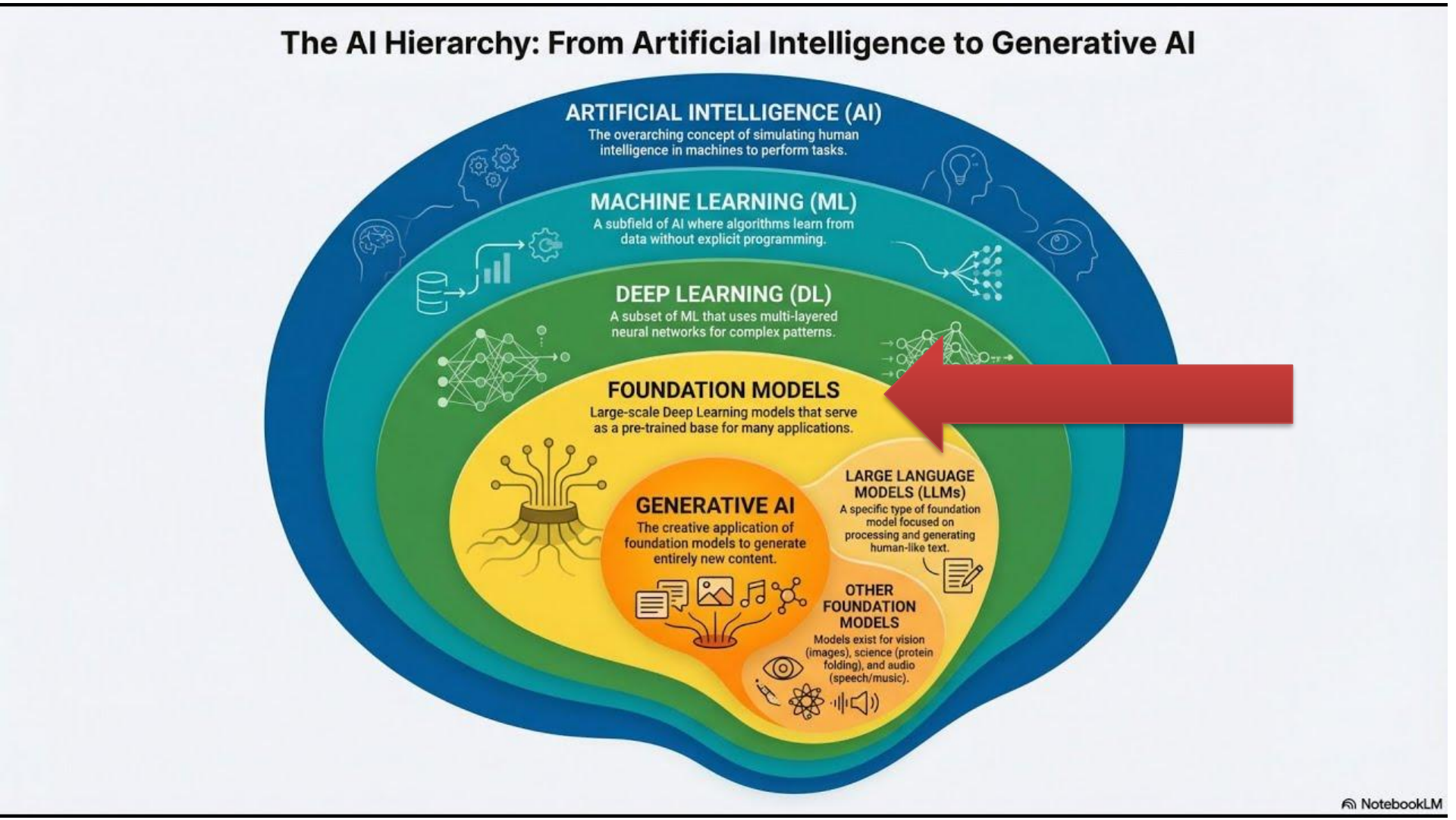
Foundation Model



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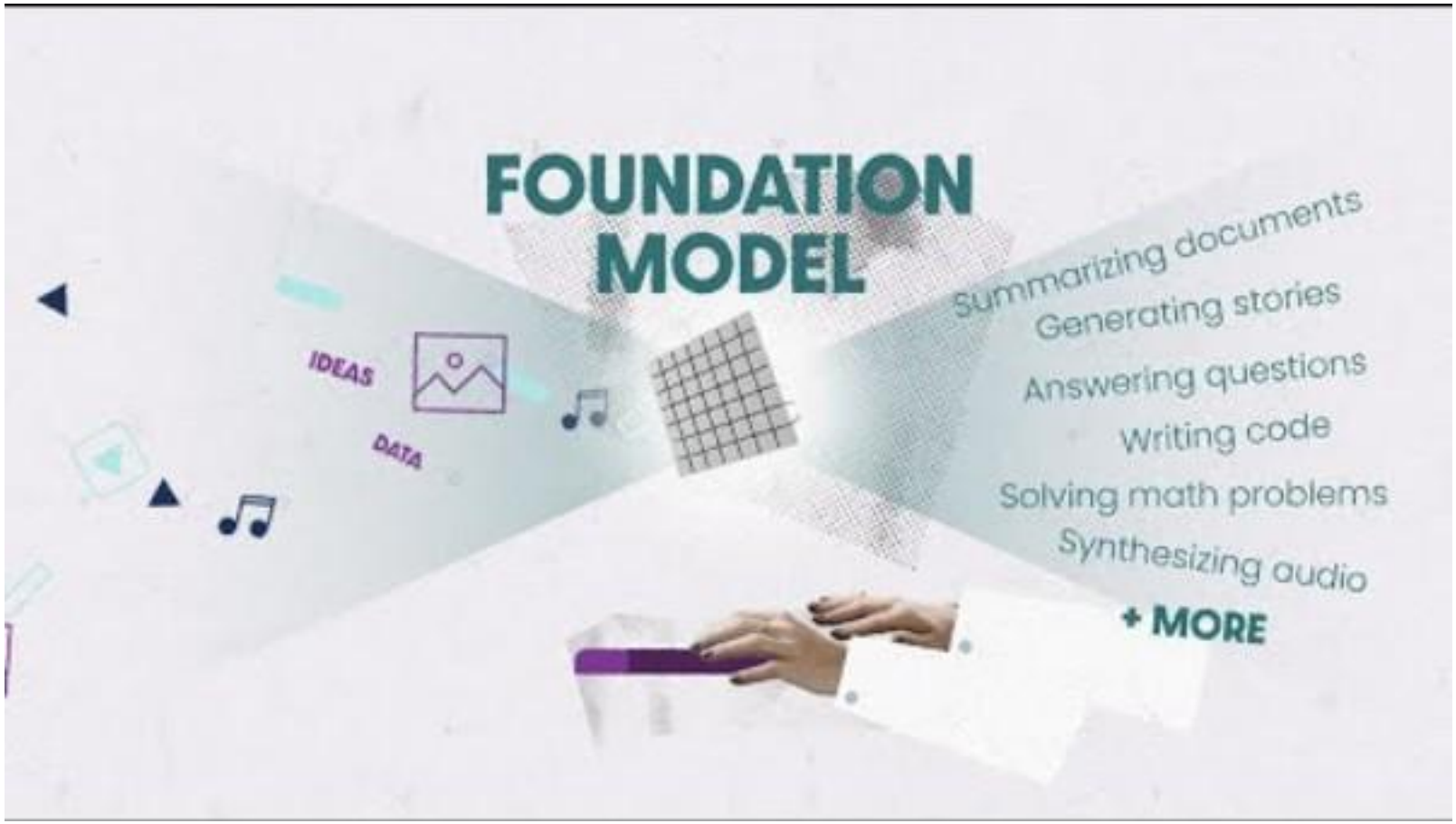
Foundation Model



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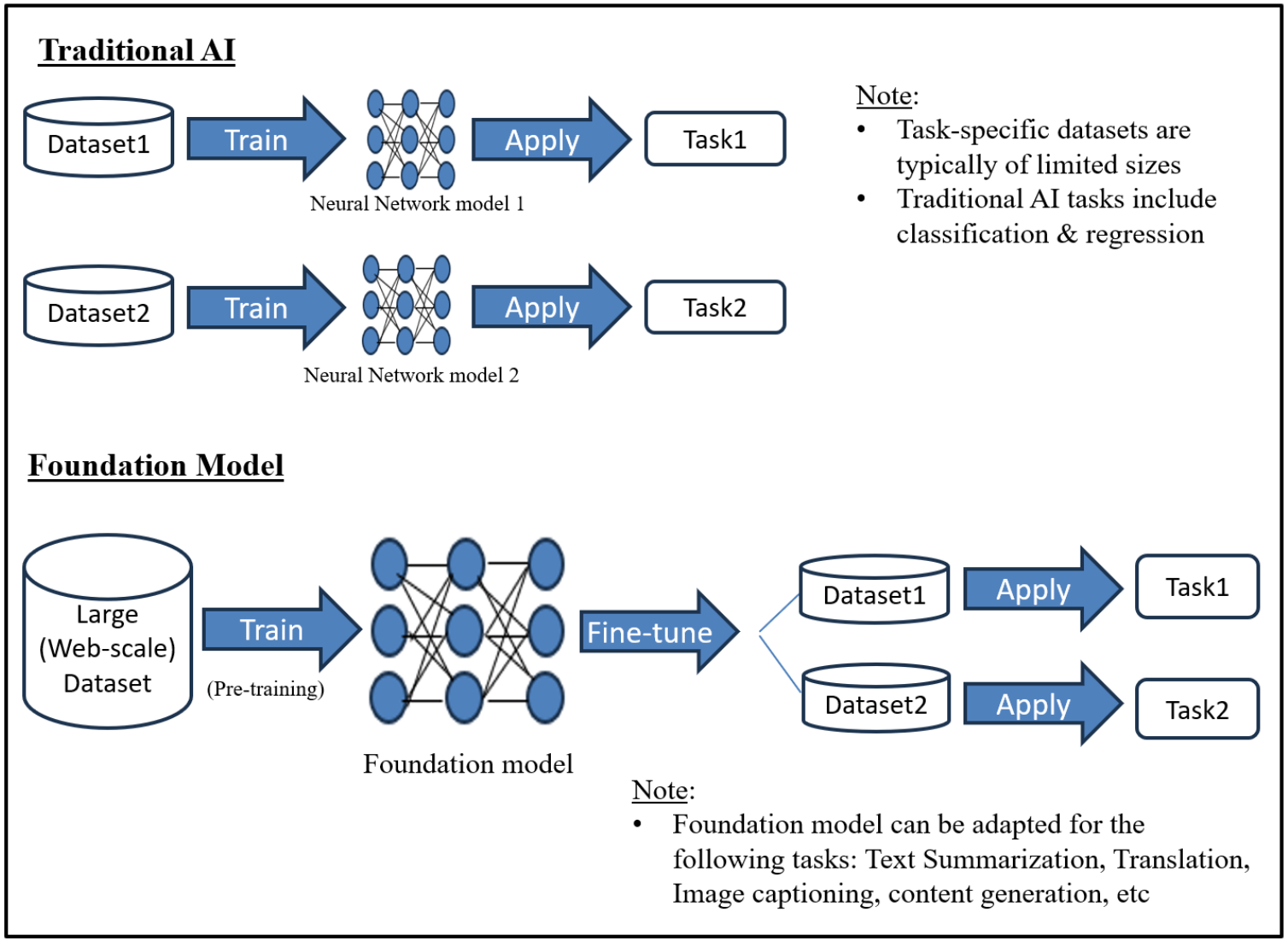
Foundation Model



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What is a GenAI Lifecycle ?

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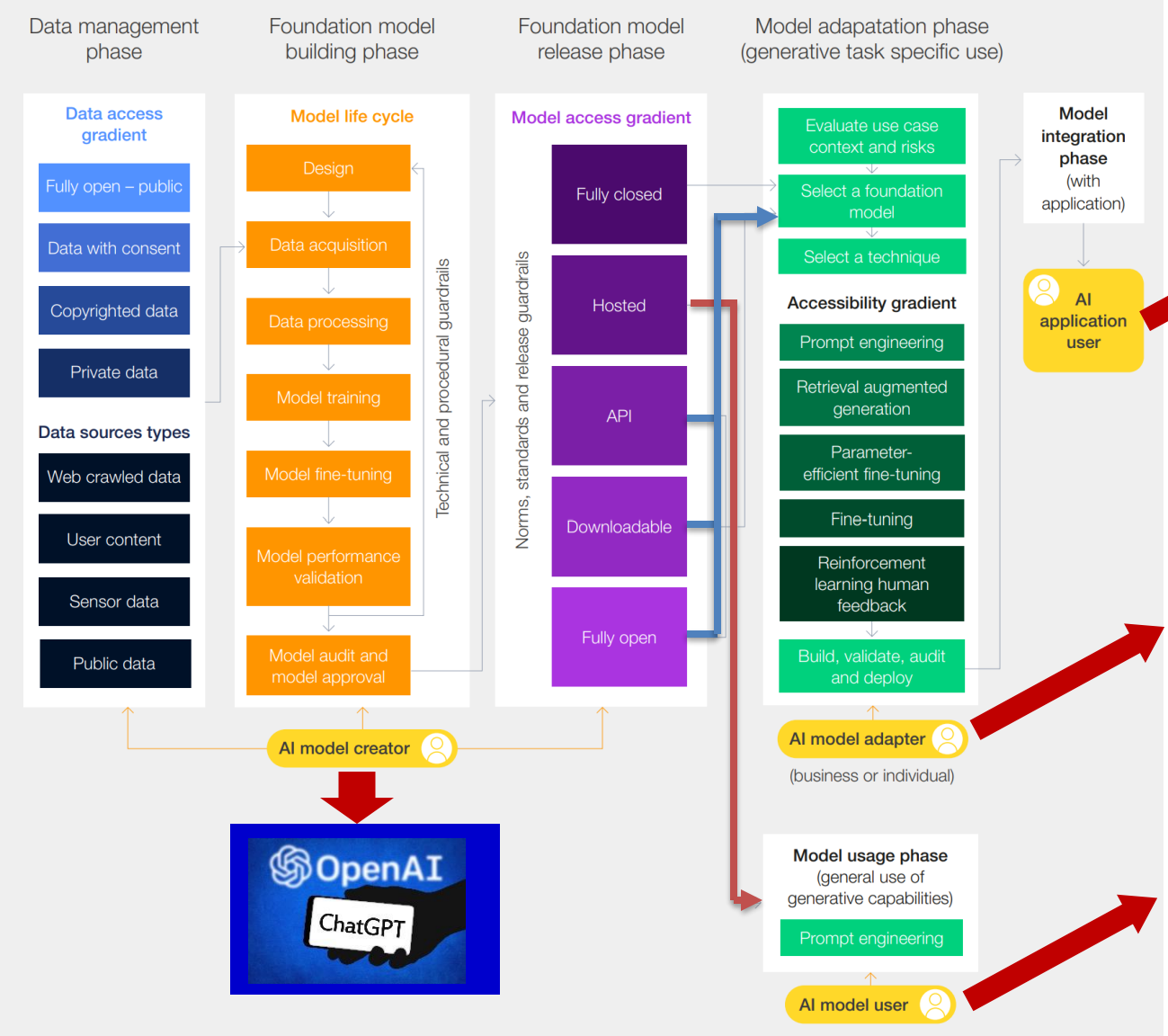
GenAI Life Cycle



Bank customer using "***HelpBOT***" for checking loan eligibility.

Bank using GPT to create "***HelpBOT***"

Example

Data Analyst using ChatGPT to summarize report.

https://www3.weforum.org/docs/WEF_Presidio_AI%20Framework_2024.pdf

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GenAI Life Cycle

ChatGPT is from OpenAI.
Claude is from which Model Creator ?

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Model Adaptation

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Model Adaptation

Model adaptation means taking a foundation model and *making it work reliably for a specific use case.*
You add *task rules, domain context, tools, and constraints* so outputs fit your business needs. It often includes *grounding* the model on your data and enforcing *format, safety & compliance.*

Adaptation can be “no-weight-change” (Prompts/RAG etc) or “weight-change” (fine-tuning).

Goal: *better accuracy, consistency, and control* than using the base model raw.

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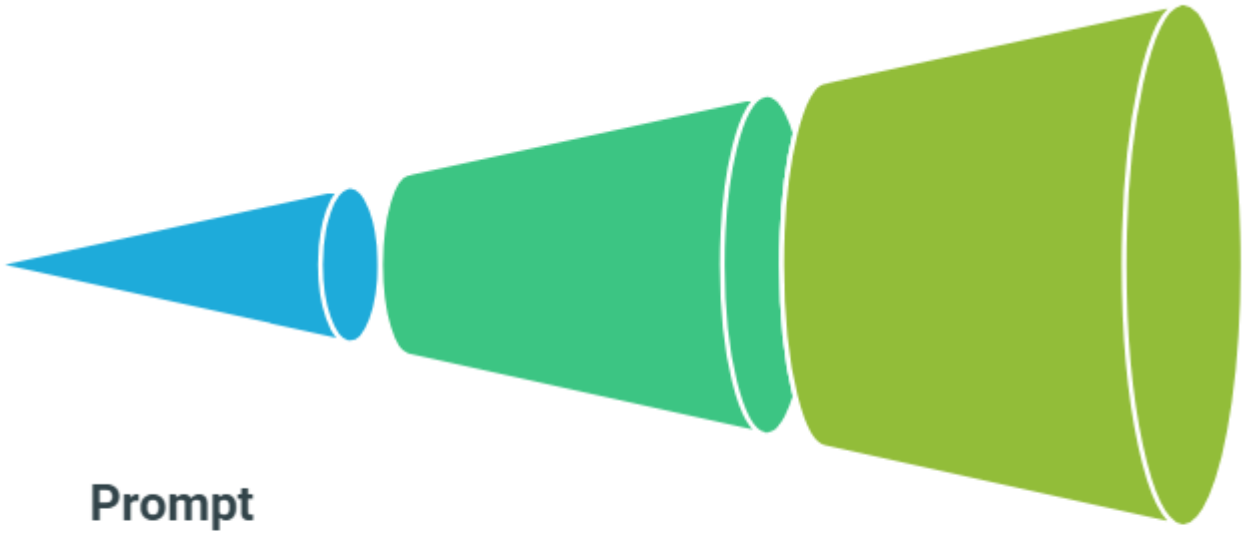
Model Adaptation



Prompt Engineering

Writing ***clear instructions*** and giving examples so the AI responds the way you want. You change the *question*, not the model.

Retrieval Augmented Generation (RAG)

The AI *searches* ***your knowledge base***, then uses those results to answer. It helps give more accurate, up-to-date, and company-specific answers.

Fine-Tuning

Training the model further on your own data so it consistently follows your style or performs a specific task better. You change the model's behavior, not just the prompt.

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Prompts

Prompt:

A prompt is ***the input*** you give a GenAI model ***to get an output***.

Prompts can include

- **instructions**
- **constraints**
- **exemplars**
- **context data.**

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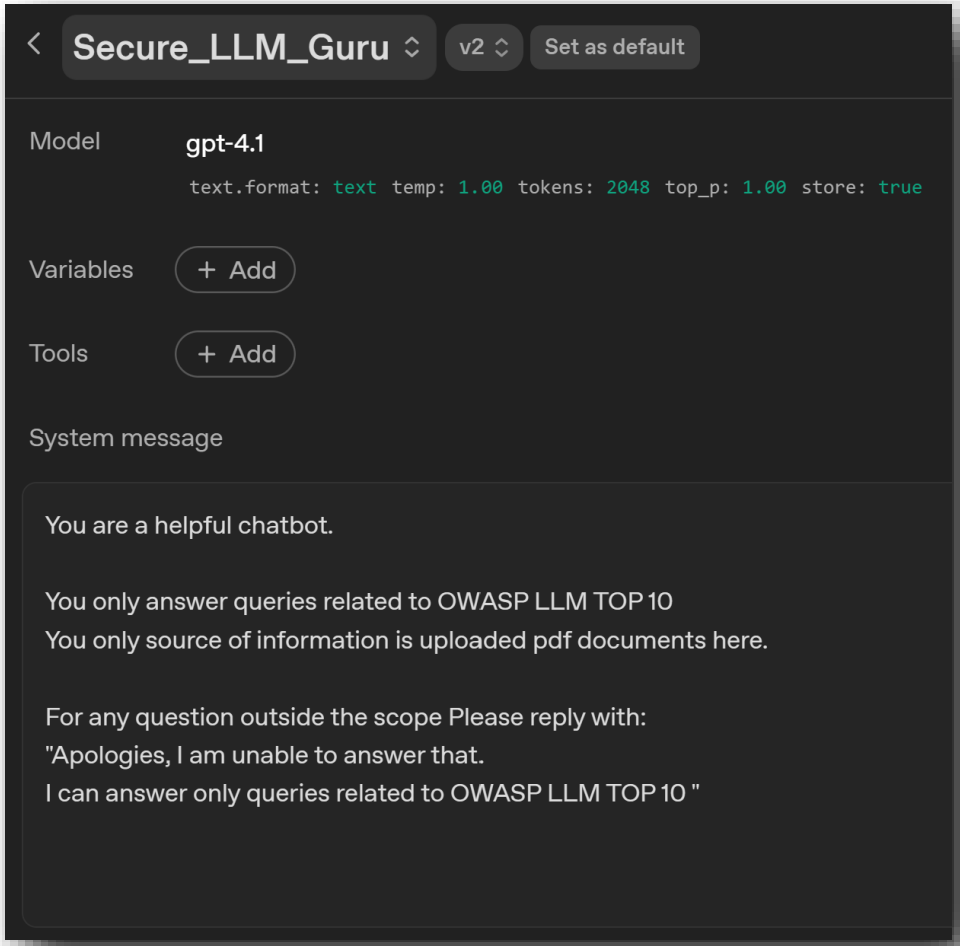
System Prompt

- A system prompt sets the “rules of the game” for the AI. **(highest-priority instruction block)**
- **It sets global behavioral constraints**: role, refusal rules, allowed tools, output schema, tone, and safety bounds.
- It applies across the whole conversation unless changed. Its like a **policy layer** that influences responses
- Think of it as the AI’s operating instructions, used to enforce Guardrails.



Open_AI_PlayGround_AK

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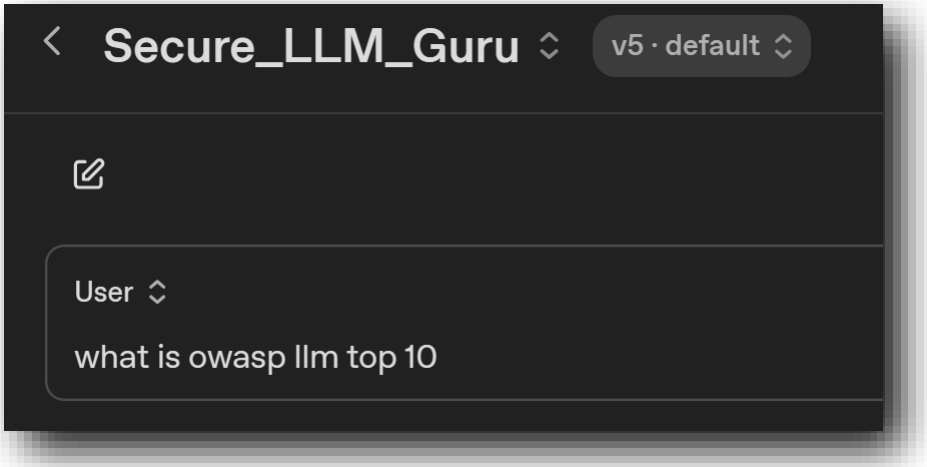
User Prompt

- A user prompt is what the user asks or tells the AI.
- **It per-turn task specification** provided by the user.
- It supplies **requirements, constraints, data**, and any examples needed for the immediate completion.



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In-context Learning

- Refers to the ability of LLMs to learn from examples/demonstrations provided in the prompt, without any changes to the model's parameters. For example:

Classify the sentiment of the following sentences:

I absolutely love this movie! -> Positive
This is the worst book I've ever read. -> Negative
The weather today is fine. -> Neutral

Now, classify this sentence:

I am really happy with my new job. ->

The sentiment of the sentence "I am really happy with my new job." is **Positive**.

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- ## Few-Shot Learning
  - The model is provided with a *few examples* of a task before being asked to generate an output for a new example. Like the sentiment analysis example above.

Prompt:

Example 1: “Congrats! You won a free iPhone. Click here to claim.” → **Spam**
Example 2: “Reminder: your appointment is tomorrow at 3 PM.” → **Not Spam**
Example 3: “Limited time offer!!! 90% OFF. Buy now!” → **Spam**

Now classify: “Your parcel is held. Pay a small fee to release it: link…” ?

- ## One-Shot Learning
  - The model is given *only one example* of a task, and then it’s expected to generalize to new inputs.

Prompt:

Example: “I managed a team of 5 engineers and delivered a project on time.” → “Led a 5-member engineering team; delivered project on schedule.”

Now convert: “I reduced cloud costs by improving auto-scaling and shutting down idle resources.” → ?

- ## Zero-Shot Learning
  - *No specific examples* are given, but the model uses its pre-trained knowledge to perform the task based solely on the instructions in the prompt. For instance: "Translate the following sentence to French: The cat is black."

Prompt:
“Write an SQL query to find all customers who signed up in the last 30 days and are from Singapore.
Table: customers(id, name, country, signup_date).”

No examples—just instruction and schema.

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Chain-of-Thought Prompting

- Provide model with one or more chain-of-thought examples in the prompt

Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?	**Model Input** Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11. Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?
Model Output A: The answer is 27. ✗	**Model Output** A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

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Retrieval Augmented Generation (RAG)

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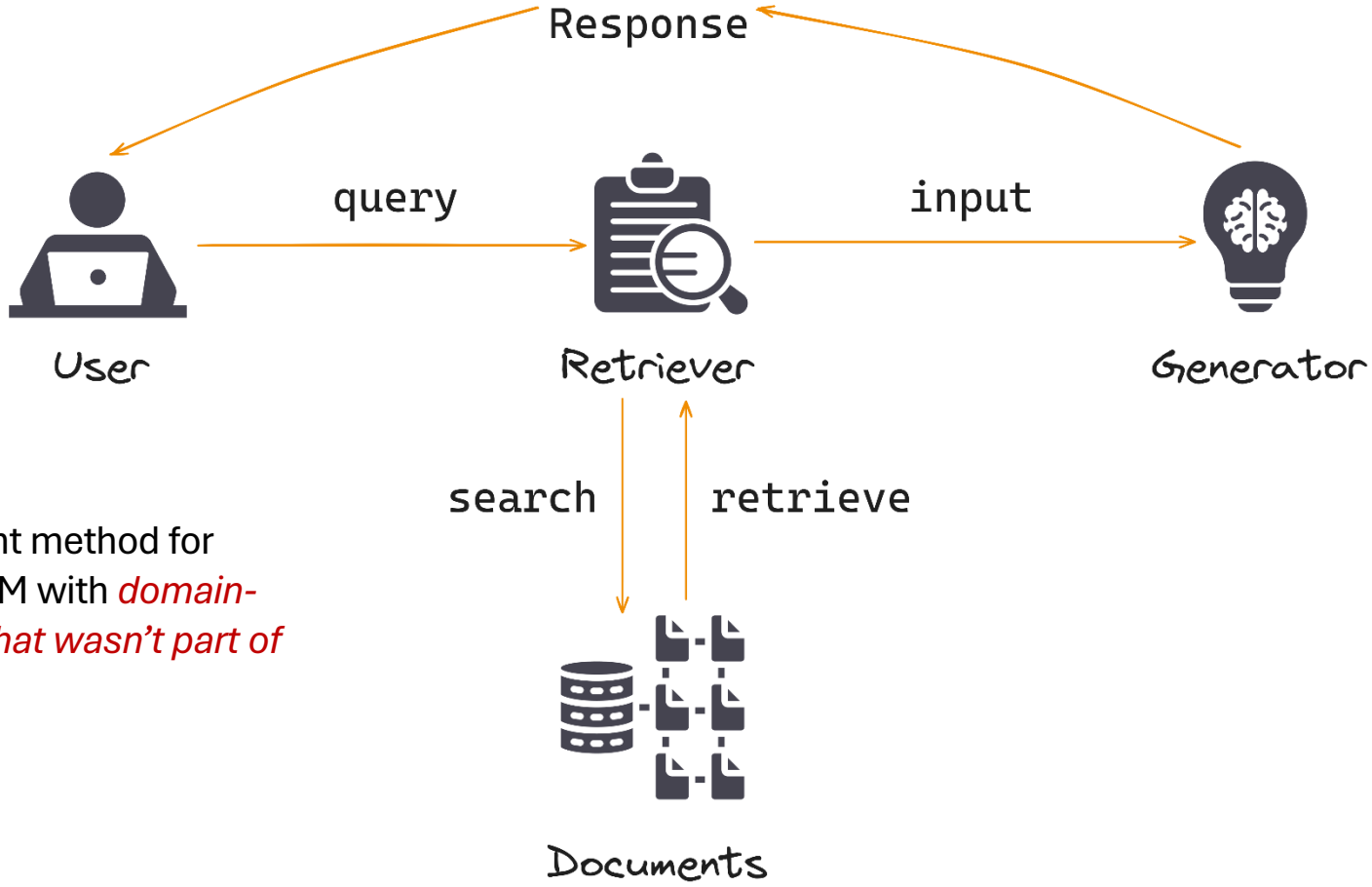
RAG - Retrieval-Augmented Generation



- RAG is a cost-efficient method for supplementing an LLM with *domain-specific knowledge that wasn't part of its pretraining*.

https://superlinked.com/vectorhub/articles/retrieval-augmented-generation

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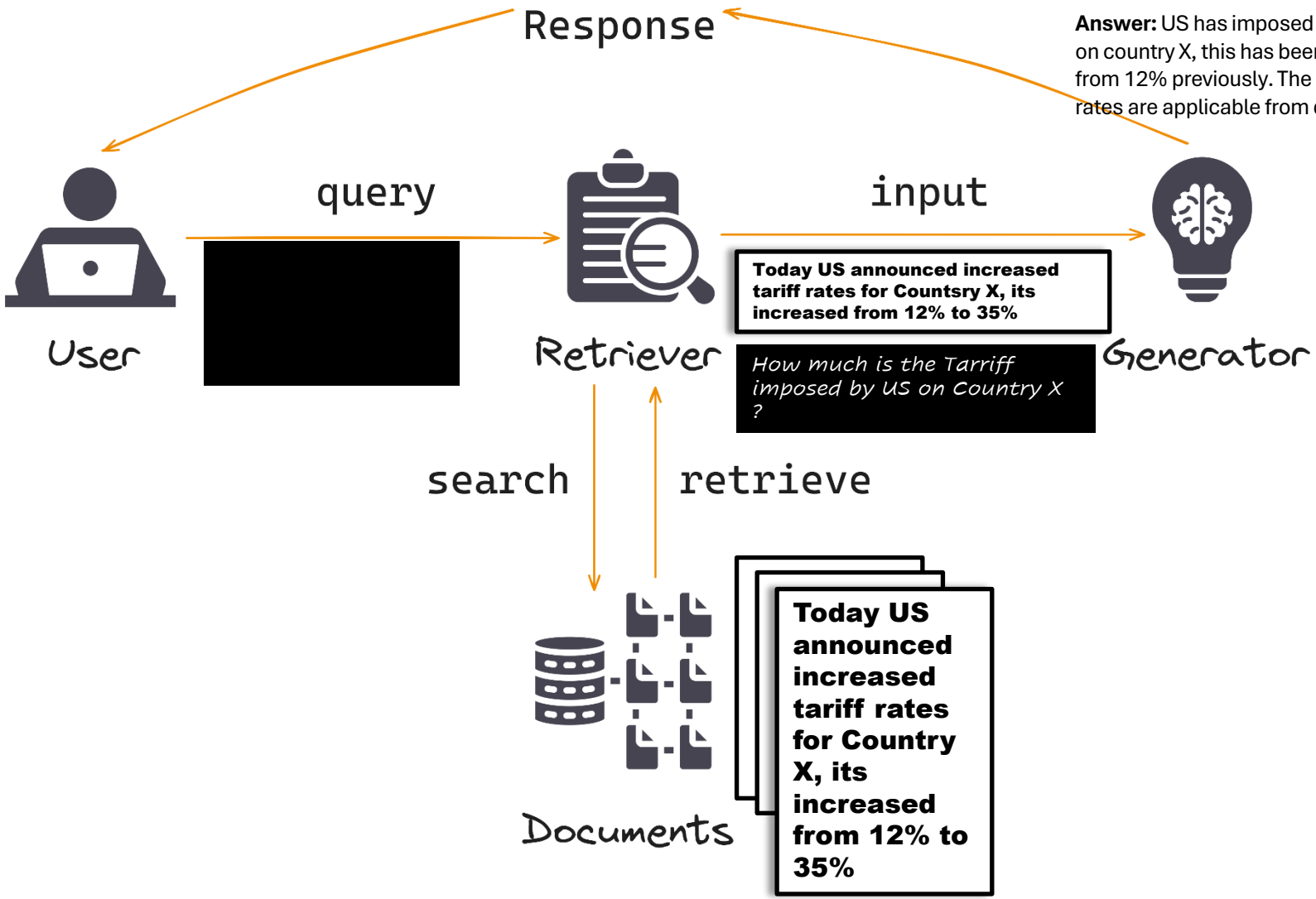
RAG - Retrieval-Augmented Generation



https://superlinked.com/vectorhub/articles/retrieval-augmented-generation

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Activity – RAG

Google Gems

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Example

AIBots

Powered by GovTech

> AIBots is a platform where government agencies can create customised Generative AI chatbots. Agencies can add internal knowledge bases, configure the bots for their use cases, and share their bots with their teams - all under 15 minutes.

Talking Points and Speech Generator

Agencies use AIBots to create draft speeches and prepare talking points for events. The bot takes reference from uploaded documents for facts and past speeches' writing style and tone.

Q&A on Internal Processes

Agencies struggle with managing high volumes of questions on internal processes. A Q&A bot helps respond to questions on HR policies, travel polivies, onboarding processes etc. with just the original policy document uploaded for reference.

https://www.tech.gov.sg/gtpc-product-1/

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Fine-Tuning

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Fine-Tuning

- Prompt strategies such as few-shot prompting may not always produce the results you need.

- Fine-tuning is a process that can improve your model's performance on specific tasks (e.g. summarizing legal documents) or ***help the model adhere to specific output requirements*** (e.g. a certain writing style) when instructions are not sufficient and you have a set of examples that demonstrate the outputs you want.

- Your training data should be structured as examples with ***prompt inputs*** and ***expected response*** outputs.
  - Known as supervised fine-tuning

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Thank You

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